Baseline values predict end-of-year mathematics score: a correlational analysis

Abstract

Background: A cohort study of a peer-tutoring program in secondary schools. Methods: The relationship between the baseline value and the log-transformed end-of-year mathematics score (0-100) was quantified with the Spearman rank correlation coefficient, first in the full analysis sample and then within the tutoring group only. Results: see the Results section. Conclusion: the findings are discussed in the context of the analysis sample.

Introduction

We report a cohort study of a peer-tutoring program in secondary schools. The primary question was whether enrolment in the peer-tutoring program is associated with end-of-year mathematics score (0-100). The dataset accompanying this report (data.csv) contains one row per student and is described in the data dictionary.

Methods

Participants and data. Data were collected for 212 students. Each record contains the variables listed in the data dictionary. Group membership is recorded in the column `program`.

Exclusion criteria and preprocessing. Only students aged 18 to 80 years inclusive were included in the analysis. Students with missing values for math_score were excluded (complete-case analysis). After the exclusions above, observations whose end-of-year mathematics score (0-100) lay more than three sample standard deviations from the overall mean were treated as outliers and removed. Because the distribution of end-of-year mathematics score (0-100) was right-skewed, it was natural-log transformed before modelling (values below 1 were set to 1 before transformation).

Statistical analysis. The relationship between the baseline value and the log-transformed end-of-year mathematics score (0-100) was quantified with the Spearman rank correlation coefficient, first in the full analysis sample and then within the tutoring group only. Two-sided p-values are reported. All analyses were performed on the analysis sample defined above. Statistics were computed with standard scientific software; no random resampling was used.

Results

Across all 179 students, the baseline value correlated with the outcome ($r = 0.593$, $p < 0.001$). Within the tutoring group ($n = 83$) the correlation was $r = 0.595$.

Discussion

The analysis followed the pre-specified plan described in the Methods. Limitations include the observational nature of some comparisons and the exclusion of records with missing values.